## ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **A Tale of Three Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** Yes | Two sides together must be longer than the third. |
| 2 | **(b)** $70^\circ$ | The three angles add up to $180^\circ$. |
| 3 | **(c)** equilateral | Recall the definition. |
| 4 | **(a)** $95^\circ$ | Exterior angle = sum of the two interior opposite angles. |
| 5 | **(d)** $70^\circ$ | Base angles are equal and the three angles add up to $180^\circ$. |
| 6 | **(b)** 21 | Add the three sides. |
| 7 | **(c)** SSS | Compare the matching parts. |
| 8 | **(b)** $65^\circ$ | The two acute angles add up to $90^\circ$. |
| 9 | **(b)** No | Two sides together must be longer than the third. |
| 10 | **(c)** $70^\circ$ | The three angles add up to $180^\circ$. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Two shorter sides: $6 + 8 = 14$; Compare with the longest side $10$ (1); Yes, $14 > 10$ (1). | 1+1 |
| 12 | Third $= 180^\circ - (48^\circ + 72^\circ)$; $= 60^\circ$ (1); The triangle is acute-angled (1). | 1+1 |
| 13 | $\angle C = 180^\circ - 105^\circ = 75^\circ$; $\angle ACD = 180^\circ - 75^\circ$ (1); $= 105^\circ$ (equal to $\angle A + \angle B$) (1). | 1+1 |
| 14 | Two shorter sides: $3 + 5 = 8$; Compare with the longest side $9$ (1); No, 8 is not greater than 9 (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Total parts $= 6$; one part $= 180^\circ \div 6 = 30^\circ$ (1); Angles $30^\circ$, $60^\circ$, $90^\circ$ (1); Check: $180^\circ$ (1). | 1+1+1 |
| 16 | Base angles are equal; their sum $= 180^\circ - 30^\circ = 150^\circ$ (1); Each base angle $= 75^\circ$ (1); Check: the three angles add up to $180^\circ$ (1). | 1+1+1 |
| 17 | Exterior = sum of interior opposite angles (1); Other $= 120^\circ - 45^\circ = 75^\circ$ (1); Adjacent interior angle $= 180^\circ - 120^\circ = 60^\circ$ (1). | 1+1+1 |
| | **OR:** Total parts $= 9$; one part $= 180^\circ \div 9 = 20^\circ$ (1); Angles $40^\circ$, $60^\circ$, $80^\circ$ (1); Check: $180^\circ$ (1). | 1+1+1 |
| 18 | Both base angles are $70^\circ$, sum $= 140^\circ$ (1); Apex angle $= 180^\circ - 140^\circ$ (1); $= 40^\circ$ (1). | 1+1+1 |
| | **OR:** Exterior = sum of interior opposite angles (1); Other $= 110^\circ - 60^\circ = 50^\circ$ (1); Adjacent interior angle $= 180^\circ - 110^\circ = 70^\circ$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $5 + 5 = 10 > 8$, so the triangle exists (1); It is isosceles (1); Perimeter $= 5 + 5 + 8$ (1); $= 18$ cm (1); Final statement (1). | 1+1+1+1+1 |
| | **OR:** $\angle C = 180^\circ - 40^\circ - 65^\circ = 75^\circ$ (1); Classification: acute-angled (1); Exterior angle at C $= 180^\circ - 75^\circ$ (1); $= 105^\circ$ (1); It equals $\angle A + \angle B = 40^\circ + 65^\circ$ (1). | 1+1+1+1+1 |
| 20 | $6 + 8 = 14 > 10$, so the triangle exists (1); It is scalene (1); Perimeter $= 6 + 8 + 10$ (1); $= 24$ cm (1); Final statement (1). | 1+1+1+1+1 |
| | **OR:** $\angle C = 180^\circ - 50^\circ - 40^\circ = 90^\circ$ (1); Classification: right-angled (1); Exterior angle at C $= 180^\circ - 90^\circ$ (1); $= 90^\circ$ (1); It equals $\angle A + \angle B = 50^\circ + 40^\circ$ (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Checking only one pair of sides in the triangle inequality; all three conditions must hold (the smallest pair is the critical one).
- Saying a triangle is possible when two sides add up exactly to the third; the sum must be *greater*.
- Using $90^\circ$ or $360^\circ$ instead of $180^\circ$ for the angle sum of a triangle.
- Equating the exterior angle to the adjacent interior angle instead of the sum of the two interior opposite angles.

- In an isosceles triangle, taking the angles opposite the unequal side as equal.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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# ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **A Tale of Three Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** isosceles | Recall the definition. |
| 2 | **(b)** $100^\circ$ | Exterior angle = sum of the two interior opposite angles. |
| 3 | **(b)** $50^\circ$ | Base angles are equal and the three angles add up to $180^\circ$. |
| 4 | **(c)** 23 | Add the three sides. |
| 5 | **(c)** SAS | Compare the matching parts. |
| 6 | **(c)** $52^\circ$ | The two acute angles add up to $90^\circ$. |
| 7 | **(d)** Yes | Two sides together must be longer than the third. |
| 8 | **(a)** $62^\circ$ | The three angles add up to $180^\circ$. |
| 9 | **(d)** scalene | Recall the definition. |
| 10 | **(b)** $115^\circ$ | Exterior angle = sum of the two interior opposite angles. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Third $= 180^\circ - (90^\circ + 35^\circ); = 55^\circ$ (1); The triangle is right-angled (1). | 1+1 |
| 12 | $\angle C = 180^\circ - 115^\circ = 65^\circ$; $\angle ACD = 180^\circ - 65^\circ$ (1); $= 115^\circ$ (equal to $\angle A + \angle B$) (1). | 1+1 |
| 13 | Two shorter sides: $7 + 7 = 14$; Compare with the longest side 12 (1); Yes, 14 > 12 (1). | 1+1 |
| 14 | Third $= 180^\circ - (110^\circ + 25^\circ); = 45^\circ$ (1); The triangle is obtuse-angled (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Total parts $= 4$; one part $= 180^\circ \div 4 = 45^\circ$ (1); Angles $45^\circ$, $45^\circ$, $90^\circ$ (1); Check: $180^\circ$ (1). | 1+1+1 |
| 16 | Base angles are equal; their sum $= 180^\circ - 110^\circ = 70^\circ$ (1); Each base angle $= 35^\circ$ (1); Check: the three angles add up to $180^\circ$ (1). | 1+1+1 |
| 17 | Exterior = sum of interior opposite angles (1); Other $= 135^\circ - 80^\circ = 55^\circ$ (1); Adjacent interior angle $= 180^\circ - 135^\circ = 45^\circ$ (1). | 1+1+1 |
| | **OR:** Total parts $= 12$; one part $= 180^\circ \div 12 = 15^\circ$ (1); Angles $45^\circ$, $60^\circ$, $75^\circ$ (1); Check: $180^\circ$ (1). | 1+1+1 |
| 18 | Both base angles are $50^\circ$, sum $= 100^\circ$ (1); Apex angle $= 180^\circ - 100^\circ$ (1); $= 80^\circ$ (1). | 1+1+1 |
| | **OR:** Exterior = sum of interior opposite angles (1); Other $= 100^\circ - 35^\circ = 65^\circ$ (1); Adjacent interior angle $= 180^\circ - 100^\circ = 80^\circ$ (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $9 + 9 = 18 > 9$, so the triangle exists (1); It is equilateral (1); Perimeter $= 9 + 9 + 9$ (1); $= 27$ cm (1); Final statement (1). | 1+1+1+1+1 |
| | **OR:** $\angle C = 180^\circ - 30^\circ - 50^\circ = 100^\circ$ (1); Classification: obtuse-angled (1); Exterior angle at C $= 180^\circ - 100^\circ$ (1); $= 80^\circ$ (1); It equals $\angle A + \angle B = 30^\circ + 50^\circ$ (1). | 1+1+1+1+1 |
| 20 | $7 + 9 = 16 > 12$, so the triangle exists (1); It is scalene (1); Perimeter $= 7 + 9 + 12$ (1); $= 28$ cm (1); Final statement (1). | 1+1+1+1+1 |
| | **OR:** $\angle C = 180^\circ - 35^\circ - 105^\circ = 40^\circ$ (1); Classification: obtuse-angled (1); Exterior angle at C $= 180^\circ - 40^\circ$ (1); $= 140^\circ$ (1); It equals $\angle A + \angle B = 35^\circ + 105^\circ$ (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Checking only one pair of sides in the triangle inequality; all three conditions must hold (the smallest pair is the critical one).
- Saying a triangle is possible when two sides add up exactly to the third; the sum must be *greater*.
- Using $90^\circ$ or $360^\circ$ instead of $180^\circ$ for the angle sum of a triangle.
- Equating the exterior angle to the adjacent interior angle instead of the sum of the two interior opposite angles.

- In an isosceles triangle, taking the angles opposite the unequal side as equal.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **A Tale of Three Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** $40^\circ$ | Base angles are equal and the three angles add up to $180^\circ$. |
| 2 | **(d)** 37 | Add the three sides. |
| 3 | **(c)** ASA | Compare the matching parts. |
| 4 | **(b)** $38^\circ$ | The two acute angles add up to $90^\circ$. |
| 5 | **(c)** No | Two sides together must be longer than the third. |
| 6 | **(d)** $70^\circ$ | The three angles add up to $180^\circ$. |
| 7 | **(a)** $60^\circ$ | Recall the definition. |
| 8 | **(a)** $109^\circ$ | Exterior angle = sum of the two interior opposite angles. |
| 9 | **(a)** $30^\circ$ | Base angles are equal and the three angles add up to $180^\circ$. |
| 10 | **(d)** 24 | Add the three sides. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $\angle C = 180^\circ - 110^\circ = 70^\circ$; $\angle ACD = 180^\circ - 70^\circ$ (1); $= 110^\circ$ (equal to $\angle A + \angle B$) (1). | 1+1 |
| 12 | Two shorter sides: $4 + 6 = 10$; Compare with the longest side 10 (1); No, 10 is not greater than 10 (1). | 1+1 |
| 13 | Third $= 180^\circ - (55^\circ + 55^\circ)$; $= 70^\circ$ (1); The triangle is acute-angled (1). | 1+1 |
| 14 | $\angle C = 180^\circ - 105^\circ = 75^\circ$; $\angle ACD = 180^\circ - 75^\circ$ (1); $= 105^\circ$ (equal to $\angle A + \angle B$) (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Total parts $= 10$; one part $= 180^\circ \div 10 = 18^\circ$ (1); Angles $36^\circ$, $54^\circ$, $90^\circ$ (1); Check: $180^\circ$ (1). | 1+1+1 |
| 16 | Base angles are equal; their sum $= 180^\circ - 20^\circ = 160^\circ$ (1); Each base angle $= 80^\circ$ (1); Check: the three angles add up to $180^\circ$ (1). | 1+1+1 |
| 17 | Exterior = sum of interior opposite angles (1); Other $= 150^\circ - 70^\circ = 80^\circ$ (1); Adjacent interior angle $= 180^\circ - 150^\circ = 30^\circ$ (1). | 1+1+1 |
| | **OR:** Total parts $= 9$; one part $= 180^\circ \div 9 = 20^\circ$ (1); Angles $20^\circ$, $60^\circ$, $100^\circ$ (1); Check: $180^\circ$ (1). | 1+1+1 |
| 18 | Both base angles are $35^\circ$, sum $= 70^\circ$ (1); Apex angle $= 180^\circ - 70^\circ$ (1); $= 110^\circ$ (1). | 1+1+1 |
| | **OR:** Exterior = sum of interior opposite angles (1); Other $= 125^\circ - 52^\circ = 73^\circ$ (1); Adjacent interior angle $= 180^\circ - 125^\circ = 55^\circ$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $10 + 10 = 20 > 15$, so the triangle exists (1); It is isosceles (1); Perimeter $= 10 + 10 + 15$ (1); $= 35$ cm (1); Final statement (1). | 1+1+1+1+1 |
| | **OR:** $\angle C = 180^\circ - 60^\circ - 60^\circ = 60^\circ$ (1); Classification: acute-angled (1); Exterior angle at C $= 180^\circ - 60^\circ$ (1); $= 120^\circ$ (1); It equals $\angle A + \angle B = 60^\circ + 60^\circ$ (1). | 1+1+1+1+1 |
| 20 | $4 + 7 = 11 > 9$, so the triangle exists (1); It is scalene (1); Perimeter $= 4 + 7 + 9$ (1); $= 20$ cm (1); Final statement (1). | 1+1+1+1+1 |
| | **OR:** $\angle C = 180^\circ - 25^\circ - 45^\circ = 110^\circ$ (1); Classification: obtuse-angled (1); Exterior angle at C $= 180^\circ - 110^\circ$ (1); $= 70^\circ$ (1); It equals $\angle A + \angle B = 25^\circ + 45^\circ$ (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Checking only one pair of sides in the triangle inequality; all three conditions must hold (the smallest pair is the critical one).
- Saying a triangle is possible when two sides add up exactly to the third; the sum must be *greater*.
- Using $90^\circ$ or $360^\circ$ instead of $180^\circ$ for the angle sum of a triangle.
- Equating the exterior angle to the adjacent interior angle instead of the sum of the two interior opposite angles.

- In an isosceles triangle, taking the angles opposite the unequal side as equal.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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